Eprad eCNA Cinema Automation Plugin

Version 1.4 | Build 32

Use this plugin to monitor and control an Eprad eCNA Cinema Automation system from within your Q-SYS designs. The plugin communicates over two independent channels: the CAI (Cinema Automation Interface) ASCII TCP protocol for commands and status, and a direct HTTP connection for I/O flag detail and, optionally, Eprad QDC-400 dimmer lighting and Eprad nanoHost POS management data.

What's New in 1.4

Version 1.4 adds optional Event Logging to the Q-SYS Core Manager, a more reliable automatic reconnection sequence for eCNA units that are slow to come up after a reboot or power-on, plus a couple of small unrelated improvements:

- **Event Logging** — new optional Event Log page that sends selected state changes (Connection, Inputs, Outputs, Program, Macros, RDI, Dimmer Control, NanoHost) to the Q-SYS Core Manager Event Log, with a master enable, a per-category enable for each, and an on-page history of the last 10 entries actually sent. See Event Logging.
- **Digital Projector alternate Event Log names** — four new text fields let you give the Dig Proj1/2 Power/Video flags a different name in Event Log messages than the fixed labels used elsewhere in the plugin. See Event Logging.
- **Improved reconnections** — the automatic reconnect sequence is substantially more reliable, especially for eCNA units that are slow to come back up after a reboot or power-on. No configuration changes are needed to get this improvement.
- **RST Poll Interval default lowered** — the default changed from 45 seconds to 5 seconds, so Segment Name/Segment Index on the Program page reflect the eCNA up to 9x faster out of the box. Existing designs pick up the faster default automatically unless this property was explicitly set otherwise. See RST Poll Interval.

Compatibility

This plugin was developed and tested with Q-SYS Designer Software version 9.13.1 LTS, and is intended to be compatible with 9.4.8 LTS and newer. It is also being tested against 9.13.2 LTS. It has been tested against Eprad eCNA firmware v1.23 and v1.26, running on both the eCNA-5 and eCNA-10.

> **Note:** The plugin communicates via the CAI TCP interface (default port 13002) and the eCNA HTTP web interface (port 80). Both must be reachable from the Q-SYS Core on your network. Because the eCNA supports only one CAI TCP connection per port at a time, avoid using the same port number in multiple concurrent designs.

> **Note:** nanoHost POS Management Data requires a physical Eprad nanoHost Cinema Integration System connected to the eCNA — it has no effect if one isn't present. See nanoHost POS Management Data.

> **Note:** Event Logging (new in 1.4) requires nothing beyond the plugin itself, but entries only reach the Core Manager Event Log while the design is running on an actual Q-SYS Core — see Event Logging.

Configuration Overview

Connect the Plugin

1. Drag the Eprad eCNA plugin into your Q-SYS schematic.
2. In the plugin Properties, configure the number of Macros and RDI devices for your installation. If the installation includes Eprad QDC-400 dimmer boards, enable Enable QDC Lighting and set the zone/channel counts — see QDC-400 Dimmer Lighting. If the installation includes an Eprad nanoHost, enable Enable NanoHost — see nanoHost POS Management Data. If you want selected events sent to the Core Manager Event Log, enable Enable Event Log — see Event Logging.
3. Save the design to your Core and run (F5).
4. On the Setup page, enter the IP address of the eCNA and confirm the TCP Port (default 13002).

The plugin connects automatically once a valid IP address is entered. A Connection Status of **OK** confirms a successful connection.

Communication Architecture

The plugin uses two simultaneous communication channels:

- **CAI TCP (port 13002):** The primary control channel. Sends commands (EVT, OUT, RDI, MAC) and receives status responses (RST, RID) and unsolicited event reports (RPT,EVT). A full RST status re-poll is also triggered immediately whenever a relevant event report arrives, and on a regular interval (see RST Poll Interval) as a background safety net — most status updates arrive via the unsolicited events, not this timer, though a program Segment change relies on this poll exclusively. Only one device may have a port open at a time.

- **HTTP (port 80):** A secondary read-only channel for I/O flag detail, providing the state of all 44 output flags and 31 input flags, including house lights, curtain, masking, lens, sound, and aux outputs. Polled independently at the Scrape Interval; when Enable QDC Lighting is on, this same connection also carries dimmer commands and status at the separate Light Poll Interval.

> **Note:** The HTTP scrape does not consume a CAI port. Multiple Q-SYS designs can scrape the eCNA web interface simultaneously without conflict. Only the CAI TCP connection is exclusive per port.

Unsolicited Event Reports

The plugin subscribes to unsolicited event reports from the eCNA (`CFG,RPTON`) on connection. The eCNA sends `RPT,EVT` messages when key state changes occur (START, STOP, CUE, projector power changes, fire stop, etc.). The plugin triggers an immediate RST status re-poll on receipt of these events, providing near-instant UI updates without relying solely on the poll timer.

Properties

Num Macros

Integer, 0–50 (default 50). Sets the number of Macro trigger buttons available on the Macros tab. Set to 0 to hide the Macros tab entirely.

Num RDI

Integer, 0–5 (default 5). Sets the number of RDI (Remote Device Interface) devices. Each device has 16 command buttons. Set to 0 to hide the RDI tab entirely.

RST Poll Interval

Decimal, 2–55 seconds (default 5, lowered in 1.4 from 45 — see below). Interval between CAI status requests (RST), which drive the entire Program page — Control State, Stopped State, Cue Number, Program Number, Segment Name/Index, and CAI Command Access. Most of the Program page also updates immediately from the eCNA's own unsolicited event reports, but a program segment change has no corresponding event to react to and can occur purely from Macro execution with no Cue involved — so Segment Name and Segment Index specifically can lag behind the actual eCNA state by up to this full interval. Lowering this value reduces that lag.

> **Changed in 1.4:** the default dropped from 45 seconds to 5 seconds, following extensive field testing (dozens of reboot/cold-boot cycles across three Q-SYS Designer Software versions) with no adverse effect observed. Existing designs pick up the faster 5-second default automatically on upgrade unless this property was already set to something else. If you'd prefer the old behavior, set it back to 45.

> **Note:** The upper limit of 55 seconds is intentional, not arbitrary — the eCNA can treat a CAI connection that goes fully idle for 60 seconds as available to a second client, so this property is capped short of that to preserve a safety margin. Raising the property's own maximum past 55 is not supported.

Scrape Interval

Decimal, 0.5–20 seconds (default 2). Interval between HTTP requests to the eCNA web interface for I/O flag detail. This is the primary update mechanism for the output and input flag indicators (house lights, curtain, masking, lens, sound, aux outputs, and input flags), and — independently of the Light Poll Interval below — also determines how often that same connection is available to service a queued dimmer command or the periodic dimmer status poll when Enable QDC Lighting is on. Values below 2 seconds may interfere with direct operation of the eCNA device; field testing found values as low as 0.5–1 second to work without issue, if faster flag response is needed.

> **Note:** If you're upgrading from an earlier version, this property was previously called Scrape Rate and only accepted whole seconds. Any custom value you had set will reset to the default (2 seconds) and will need to be re-entered.

Light Poll Interval

Decimal, 0.5–20 seconds (default 3). Interval between HTTP requests to the eCNA for dimmer status (zone offsets, preset state, and channel levels) — see QDC-400 Dimmer Lighting. This property is always shown regardless of the Enable QDC Lighting setting (same as QDC Zones/QDC Channels), but only takes effect once that property is turned on. It shares the same HTTP connection as Scrape Interval, but runs on its own independent cadence: I/O flag scraping still happens on every available turn, and a dimmer status poll claims a turn only once its own interval has elapsed, so lowering Scrape Interval no longer indirectly slows dimmer responsiveness the way it once did. Field testing found lowering this to 3 seconds (from the same default as Scrape Interval) noticeably improved dimmer preset/offset responsiveness with no adverse effect; there is no reason to raise it above the default other than temporary testing.

> **Note:** This and Scrape Interval share one HTTP connection, so setting either one very low reduces how often the other effectively runs — there's a practical floor to how fast both can go at once.

Show Debug Tab (removed in 1.2)

> **Removed:** Its function is handled by the Q-SYS Debug window.

Enable QDC Lighting

Boolean (default off). Turns on monitoring and control of Eprad QDC-400 dimmer lighting boards, adding the Dimmer Control page and its associated controls. See QDC-400 Dimmer Lighting for full details.

QDC Zones

Integer, 1–16 (default 2). Number of lighting zones to display on the Dimmer Control page. Zone 1 is always House; Zone 2, if included, is always Stage; any zones above 2 are user-defined on the eCNA. Set to 1 for a House-only installation. Only shown in Properties once Enable QDC Lighting is turned on.

QDC Channels

Integer, 1–16 (default 2). Number of dimmer channel meters to display on the Dimmer Control page. A single QDC-400 board supports 4 channels; more than one board can be daisy-chained. Only shown in Properties once Enable QDC Lighting is turned on.

Enable NanoHost

Boolean (default off). Turns on monitoring of the Eprad nanoHost POS Management Data, adding the NanoHost page (appended after Program) and its associated controls. See nanoHost POS Management Data for full details.

> **Note:** This page and its controls only exist when this property is on. Requires a physical Eprad nanoHost device connected to the eCNA.

Enable Event Log

Boolean (default off). Turns on Event Logging to the Q-SYS Core Manager, adding the Event Log page (appended last) and its associated controls — a master enable, a per-category enable for each event type, and an on-page history of recent entries. See Event Logging for full details.

Controls

Setup Page

IP Address

Text entry. Enter the IP address of the eCNA. The plugin connects automatically when a valid address is entered or changed.

TCP Port

Text entry. The CAI TCP port number (default 13002). Change this to match the CAI channel assigned to this Q-SYS design. Common values: 13000 (channel 1), 13001 (channel 2), 13002 (channel 3), 13003 (channel 4).

> **Note:** Port 13003 (channel 4) requires eCNA firmware version 1.25 or later.

Status LED / Status

Indicator LED and text field, read only. Displays the current connection state:

- **OK** (green): Connected and communicating.
- **Initializing** (blue): Connecting or establishing initial communication.
- **Disconnected / Fault** (red): Connection lost or error received.
- **Missing** (orange): No IP address configured.

Reconnect

Trigger button. Forces an immediate reconnection attempt. Useful if the connection has stalled, the IP address was changed, or the port was changed. Note: changing the TCP Port does not take effect until Reconnect is pressed.

Model

Text field, read only. Displays the eCNA device model returned by the RID command on connection.

Firmware

Text field, read only. Displays the eCNA firmware version returned by the RID command.

Screen Id

Text field, read only. Displays the eCNA's configured screen number (0–63).

Refresh Setup Data

Trigger button. Re-fetches everything that's normally only fetched once, at connect: input flag names, output flag names, and — when Enable QDC Lighting is on — zone names, per-channel dimmer level/fade settings, and the QDC-400 board count. Use this after changing any of that data on the eCNA itself (Setup pages); day-to-day operation does not need it.

> **Note:** This data changes rarely if ever after commissioning, so it is not automatically re-polled in the background.

Outputs Page

The Outputs page contains all eCNA output controls, grouped by type. Toggle buttons both send commands to the eCNA and reflect the current output state, most of them as returned by the web scrape.

Digital Projectors (Dig Proj1 Power, Dig Proj1 Video, Dig Proj2 Power, Dig Proj2 Video)

Toggle buttons. Control and indicate the power and video output state of the two digital projectors. These update as soon as the eCNA reports the change — typically near-instant.

> **Note:** Each of these four flags can be given an alternate name shown in Event Log messages (blank = the fixed label shown here), configured on the Event Log page — see Event Logging. This only affects Event Log message text, not these button labels.

Slide Projector

Toggle button. Controls and indicates the power state of the slide (auxiliary) projector. Updates at the regular Scrape Interval.

House Lights

Mutually exclusive Toggle buttons: None / Down / Mid1 / Mid2 / Up. Sends `OUT HLxxx` commands.

Stage Lights

Mutually exclusive Toggle buttons: None / Down / Up. Sends `OUT SLxxx` commands.

Curtain

Mutually exclusive Toggle buttons: None / Close / Open. Sends `OUT CURTxxx` commands.

Masking

Mutually exclusive Toggle buttons: None / Flat / Scope / Special. Sends `OUT MSKxxx` commands.

Lens

Mutually exclusive Toggle buttons: None / Flat / Scope / Special. Sends `OUT LENxxx` commands.

Sound

Mutually exclusive Toggle buttons: None / Aux1 / Aux2 / Dig1 / Dig2 / Mono / NonSync / SR / SVA. Sends `OUT SNDxxx` commands.

Mute

Independent Toggle button. Sends `OUT MUTEON` or `OUT MUTEOFF`.

Aux Output Flags 1–16

Toggle buttons arranged in two columns of eight. Each row shows the flag number (Out 1–Out 16), the toggle button, and the user-defined name fetched from the eCNA. Sends `OUT OUTnON/OFF` commands.

> **Note:** Names refer to Output Flag names as configured in the eCNA web interface, not physical output names on any I/O board. To refresh names after changing them on the eCNA, use Refresh Setup Data on the Setup page.

Dimmer Control Page

Visible only when Enable QDC Lighting is on. Replicates the eCNA's own *Status: Light Control* web page, with animated level meters. See QDC-400 Dimmer Lighting below for full details on this page's controls and behavior.

Inputs Page

Displays all 31 eCNA input flags as LED indicators. Two columns: General Inputs (In1–In16) on the left and System Inputs on the right. Each General Input row shows the LED, the flag number (In1–In16), and the user-defined name fetched from the eCNA.

> **Note:** Names refer to Input Flag names as configured in the eCNA web interface, not physical input names on any I/O board. Input flags are read-only indicators; they cannot be set from the plugin. To refresh names after changing them on the eCNA, use Refresh Setup Data on the Setup page.

Macros Page

Visible only when Num Macros is greater than 0. Contains trigger buttons for Macros 1 through N, arranged 10 per row. Each press sends `EVT MAC,n` to the eCNA. Text labels appear above each button.

RDI Page

Visible only when Num RDI is greater than 0. Contains trigger buttons for up to 5 RDI (Remote Device Interface) devices, each with 16 commands (labelled RDI 1-1 through RDI 5-16), arranged in two rows of 8 per device. Each press sends `EVT RDI,device,command` to the eCNA. Text labels appear above each button.

Program Page

Transport Controls (Start, Stop, Cue, Abort)

Trigger buttons. Send `EVT STA`, `EVT STP`, `EVT CUE`, and `EVT ABT` commands respectively.

Fault Controls (Set Fault, Clear Fault, Reset Fault, Cancel Alarm)

Trigger buttons. Send `EVT FLT`, `EVT CLR`, `EVT RES`, and `EVT CNL` commands respectively.

Program # / Set Prog / Set & Run

Combo box and trigger buttons. Select a program number (1–9) and press Set Prog to send `EVT PGM,n`, or Set & Run to send `EVT SPR,n` (set and immediately start).

Control State

Text field, read only. Displays the current control state from the RST response: **IDL** (idle) or **RUN** (running).

Stopped State

Text field, read only. Displays the stopped state: **OK**, **STP** (stopped), **FLT** (fault), or **FIR** (fire stop active).

Cue Number

Text field, read only. Displays the current cue number within the running program.

Program Number

Text field, read only. Displays the currently active program number.

Segment Name

Text field, read only. Displays the name of the current program segment. Unlike most of the Program page, this updates only on the next RST Poll Interval cycle rather than immediately — see that property for how to shorten this lag.

Segment Index

Text field, read only. Displays the current segment index within the program. Updates on the same cadence as Segment Name above — see RST Poll Interval.

CMD Access

Text field, read only. Displays the CAI command access status: **OK** (full access) or **BLK** (blocked — the eCNA is not accepting CAI commands).

NanoHost Page

Visible only when Enable NanoHost is on. Appended after the Program page. Replicates the eCNA's own *Status: Management Data* web page. See nanoHost POS Management Data below for full details on this page's controls and behavior.

Event Log Page

Visible only when Enable Event Log is on. Appended last, after every other page. See Event Logging below for full details on this page's controls and behavior.

Input & Output Flag Names

The plugin can display user-defined names for the 16 General Input flags (In1–In16) and the 16 Aux Output flags (Out1–Out16). These names are configured on the eCNA web interface under Setup → Input Flag Names and Setup → Output Flag Names.

How Names Are Fetched

Input and output flag names are retrieved automatically from the eCNA on initial connection, and are not automatically re-fetched afterward — they very rarely change once a system is commissioned. If names are changed on the eCNA later, use Refresh Setup Data on the Setup page to pick up the change.

Manual Refresh

The Refresh Setup Data button on the Setup page immediately refreshes input and output names (along with dimmer zone/channel data, if applicable), typically appearing within a few seconds.

Where Names Appear

- **Inputs page:** Each General Input row (In1–In16) shows the LED indicator, the flag number label, and the user-defined name in a text field alongside it.
- **Outputs page:** Each Aux Output Flag row shows the flag number, the toggle button, and the user-defined name in a text field to the right of the button.
- **Control pins:** `In Names 1–16` and `Out Names 1–16` expose the names as readable output pins for use elsewhere in the Q-SYS design.

> **Note:** Names reflect the flag names as configured in the eCNA, not the physical label on any I/O board terminal. The eCNA supports up to 14 characters per name.

QDC-400 Dimmer Lighting

When Enable QDC Lighting is turned on in Properties, the plugin adds a Dimmer Control page that monitors and controls Eprad QDC-400 dimmer boards, replicating the layout and behavior of the eCNA's own *Status: Light Control* web page.

> **Note:** The CAI TCP protocol has no commands for Zone 3–16 lighting, individual dimmer channel levels, or the light level offset — only House and Stage lighting (already present on the Outputs page in v1.1) are reachable that way. All QDC-400 monitoring and control in this section is done through the eCNA's HTTP interface instead, sharing the same connection used for I/O flag detail.

Zones

Each configured zone (up to QDC Zones, set in Properties) is shown as a row with:

- **Zone name** — House and Stage (zones 1–2) are fixed; zones 3–16 are user-defined on the eCNA (Setup → Zone Names) and fetched automatically.
- **Preset buttons** — five mutually exclusive toggle buttons: Up, Down, Mid1, Mid2, None. Selecting one drives that zone's channels to the level configured for that preset on the eCNA (Setup → QDC-400 Dimmers), and resets the zone's offset to 0.
- **Offset fader and text box** — an adjustment from –100% to +100%, applied on top of the active preset's configured level for every channel assigned to that zone (clamped to 0–100%). When the active preset is None, the offset instead becomes the absolute level directly. Dragging the fader only

sends a command to the eCNA once it has been still for about 1 second, to avoid flooding the connection while dragging; the displayed value updates immediately regardless.

- **Offset increase / decrease buttons** — small +/- buttons beside the offset fader and text box. A quick press nudges the offset by 1%; press and hold for about 2 seconds to begin auto-repeating at 1% every 0.5 second until released, always clamped to –100%/+100%. Like the fader, the displayed value and the channel meter animation both update immediately on every press, but the command actually sent to the eCNA is held until input has been idle for about 1 second — so a quick burst of taps, or a long hold, results in one command for the value you land on, not one per tap.

Dimmer Channels

Each configured channel (up to QDC Channels, set in Properties) is shown as a vertical level meter with:

- **Level meter and percentage** — the channel's current output level, 0–100%.
- **Zone** — the name of the zone this channel is currently assigned to on the eCNA (Setup → QDC-400 Dimmers). More than one channel can be assigned to the same zone.
- **Present indicator** — a small LED that lights once the channel has reported a genuine non-zero level at least once. See Dimmer Presence Detection below.

Animated Level Meters

The eCNA can only be polled for dimmer status every couple of seconds at most (shared with the I/O flag scrape, which is intentionally kept fast and responsive since it is the most frequently used part of the plugin). A real dimmer fade often completes faster than that, so rather than have the meters visibly jump or lag behind, the plugin locally animates each channel toward its expected destination using the actual Level %/Fade-Time settings configured on the eCNA (Setup → QDC-400 Dimmers), the same values the physical dimmer itself uses.

This animation runs whenever a preset change is detected — whether it came from this plugin's own buttons, from the eCNA's own automation (a cinema server sending it a cue, a macro, another control system), or directly at the physical unit. House and Stage changes are typically detected within about a second via the same fast I/O flag scrape used for other outputs; other zones are detected on the next dimmer status poll.

The fade duration is the time to go from whichever preset was previously active to the newly selected one, regardless of how far apart their configured levels are — a short hop and a full 0–100% sweep at the same configured rate take about the same time. Offset adjustments always take about 3 seconds to reach their new level, regardless of which preset is active. Selecting None is effectively instant.

> **Note:** This is a prediction, not a live readout, while a channel is animating — the plugin deliberately keeps showing its own smooth estimate rather than a possibly jumpy raw reading, and simply lets the animation finish on its own schedule. It resyncs with the eCNA's actual reported level as soon as the animation completes.

Dimmer Presence Detection

Not every QDC-400 channel necessarily has a physical dimmer connected and powered — for example during commissioning, or on a system with more channels wired than are actually driving fixtures. If a channel is expected to be at a non-zero level (based on its configured preset and offset) but the eCNA reports 0% two poll cycles in a row, the plugin assumes there's no working dimmer on that channel and stops trying to animate it — showing the honest 0% reading instead of repeatedly predicting a level that never arrives. A single genuine non-zero reading, at any time, immediately clears that assumption and lights the channel's Present indicator.

As a second, independent check, the plugin also scans the eCNA's Local I/O Network (Status → Local I/O Network) for active QDC-400 boards, shown as QDC Boards Found on the Dimmer Control page. If zero active boards are found, every channel is immediately marked as having no dimmer present, without waiting on the per-channel detection above. This scan runs once on connect and otherwise only via Refresh Setup Data — it is not repolled in the background.

nanoHost POS Management Data

When Enable NanoHost is turned on in Properties, the plugin adds a NanoHost page (appended after Program) that displays live point-of-sale management data from an Eprad nanoHost Cinema Integration System, replicating the layout and behavior of the eCNA's own *Status: Management Data* web page. This page is informational only — nothing on it sends any command to the eCNA or the nanoHost.

> **Note:** This requires a physical Eprad nanoHost device connected to the eCNA. The nanoHost itself updates the data it reports no faster than every 5 minutes, so this page is polled far more slowly than the regular I/O flag scrape — see Automatic Refresh below for how the plugin keeps the displayed flags reasonably current despite that.

The nanoHost receives just the schedule information from the POS. As such, it does not know how long any preshow (ads, trailers, etc.) is. The flags and timers below change based on what the listed show times (and provided feature lengths) are.

To allow for this lack of information, the flags/states change, as outlined below, to ensure that the eCNA can accurately know when tickets have been sold with respect to when a show is in progress.

Status Flags

Six status indicators, shown as colored LEDs (green when true, red when false, except POS Cancel which uses reverse logic):

- **POS Ticket Sold** — at least one ticket has been sold for the currently-relevant performance.
- **POS In Show** — a performance is currently in progress.
- **POS Ok** — the POS system is sending schedules.
- **POS Enabled** — the POS integration is enabled on the nanoHost.
- **nanoHost Up** — the nanoHost itself is online and reachable by the eCNA. When this is off, every Show Timer and the entire Show Schedule are blanked rather than left showing stale numbers — see Automatic Refresh.
- **POS Cancel** — reverse logic: **green** means not cancelled (0), **red** means cancelled (1). This is the only one of the six flags where "on" is the concerning state.

Show Timers

Seven read-only text fields:

- **Time of Day** — a live clock generated by the Q-SYS Core itself, ticking every second. Not read from the nanoHost.
- **Feature Run Time** — the configured duration of the current/next feature.
- **Time Between Features** — the configured gap between features.
- **Time Before Feature Start** / **Time After Feature Start** — count down to, then up from, the next feature's start time.
- **Time Before Feature End** / **Time After Feature End** — count down to, then up from, the current feature's end time.

Because the nanoHost only reports fresh data every few minutes at best, the four Before/After timers are extrapolated locally, ticking once per second between actual updates, and corrected back to the real reported value every time a fresh one arrives. The "After" timer in each pair only starts counting once its matching "Before" timer has actually reached 0:00 — they can never both show a nonzero value at the same time, matching how the real nanoHost itself behaves. All four are capped at 255:59, matching the nanoHost's own maximum; a Before timer already sitting at 255:59 is treated as "no data yet" and does not count down until a real value arrives.

> **Note:** All four Before/After timers are blanked (showing `--:--`) until the first real update has been received, and again any time nanoHost Up is false — a stale-looking frozen number could otherwise be mistaken for current data. Time of Day is not affected, since it isn't sourced from the nanoHost at all.

Show Schedule

A table of up to 10 performances for the day, each showing its start time, run time, performance number, sold/cancelled status (as colored LEDs, same green/red convention as the Status Flags above), and title. Rows with no scheduled performance are hidden automatically.

Each row also shows four small marker LEDs and, next to the StartTime and RunTime fields, two highlight LEDs:

- **FS (First Show Start)** / **FE (First Show End)** — mark the day's first performance; FS stays lit until that performance ends, FE lights only while it's actually in progress.
- **LS (Last Show Start)** / **LE (Last Show End)** — the same, for the day's last performance.
- **Cyan highlight** (next to StartTime) — lights on whichever performance is currently active. The POS Ticket Sold and POS In Show Status Flags refer to this performance.
- **Yellow highlight** (next to RunTime) — lights once that performance is in progress and past its posted showtime.

The active show (Cyan highlight) is the show that is between "Time Before Feature Start" being less than 255:59 and the "Time After Feature End" being less than 255:59. This will have the first show of the day become the active feature once the "Time Before Feature Start" timer starts counting down.

Similarly, the show that is running (Yellow highlight) is set when "Time Before Feature Start" reaches 0:00 for the active feature. It will remain active until the next feature is in its show. The last show will keep its Yellow highlight active until "Time After Feature End" reaches 255:59.

> **Note:** 10 rows is a practical maximum, not a hard limit reported by the nanoHost — typical installations see 1–6 performances per day.

Show Schedule Flags

Four additional flag-type outputs turn the FS/FE/LS/LE Show Schedule markers described above into ready-to-use automation triggers, each with its own adjustable Offset so it can be tuned to the amount of preshow content (ads, trailers, etc.) actually playing ahead of the posted showtime. All four compare against the same live Time of Day clock used elsewhere on this page, so they respond immediately as each minute ticks over rather than waiting on the next nanoHost poll.

Day Start

State LED and Trigger. Turns on once Time of Day reaches the first show's StartTime plus the Day Start Offset, and stays on until FS (First Show Start) itself goes false. The Trigger fires once, at the moment Day Start turns on.

Offset range: -180 to +60 minutes. Set the Offset to the desired lead time before the first show, including your typical preshow length ahead of the posted Start Time.

Day End

State LED and Trigger. Turns on once Time of Day reaches the last show's StartTime plus RunTime plus the Day End Offset — this can land after midnight, on the following calendar day, without issue. Stays on until LE (Last Show End) itself goes false. The Trigger fires once, at the moment Day End turns on.

Offset range: -60 to +240 minutes. Set the Offset to allow for normal cooling and content transfers, plus any compensation needed for preshow time ahead of the posted Start Time.

First Show End

Trigger only — there is no state LED for this one. The Trigger fires once per day, at whichever of the following happens first: Time of Day reaches the first show's StartTime plus RunTime plus the First Show End Offset, or the FE (First Show End) marker itself turns off before that time is reached. FE stays on through the gap between the first and second shows (Time Between Features), only turning off once that gap times out — around when the second show starts. The second condition is a ceiling: if the Offset is long enough that the first condition hasn't been reached by the time FE actually turns off, the Trigger fires immediately rather than letting a large Offset push it out past the second show's start.

Offset range: -60 to +60 minutes. Set the Offset to compensate for Preshow content.

Last Show Start

State LED and Trigger. Turns on once Time of Day reaches the last show's StartTime plus the Last Show Start Offset, and stays on until LS (Last Show Start) itself goes false. The Trigger fires once, at the moment Last Show Start turns on.

Offset range: -90 to +60 minutes. Set the Offset to compensate for the Preshow content.

> **Note:** Each Trigger fires only once per day and re-arms automatically the next time its governing marker (FS, FE, or LS) resets with a new day's schedule — no manual reset is needed.

Automatic Refresh

This page is polled roughly every 2 minutes — deliberately much slower than the regular I/O flag scrape, since the nanoHost's own data doesn't update any faster than every 5 minutes regardless. In addition to that routine poll, the plugin requests an extra, early update from the nanoHost shortly after a feature's start time arrives, and again shortly after its end time arrives.

Event Logging

When Enable Event Log is turned on in Properties, the plugin adds an Event Log page (appended last) that sends selected state changes to the Q-SYS Core Manager Event Log, alongside a short on-page history of the most recent entries actually sent.

> **Note:** An entry is added to the Event Log based on the categories enabled below.

> **Note:** Each entry's timestamp reflects when the plugin received the information from the eCNA, not necessarily the exact moment the change occurred on the eCNA itself. Some latency is normal and expected, due to the polling intervals involved and the communication time between the eCNA and the plugin.

Enable and Categories

A master Enable toggle at the top of the page turns Event Logging on or off entirely; none of the category toggles below have any effect unless it's on. Each category below is independent and defaults off:

- **Connection** — logs Connection status changes, including the initial connection process (the same states shown by Status on the Setup page).
- **Inputs** — logs Input flag changes, one entry per transition.
- **Outputs** — logs Output state changes across the Outputs page (house/stage lights, curtain, masking, lens, sound, digital/slide projectors, aux outputs).
- **Program** — logs Program-page state changes: Control State, Stopped State, Program Number, Segment, and CAI-level errors.
- **Macros** — logs Macros triggered by this plugin. Macros run directly on the eCNA (from a cinema server, another control system, or the eCNA's own front panel) are not visible to the plugin and are not logged.
- **RDI** — logs RDI commands triggered by this plugin, with the same caveat as Macros above. A row of small per-device toggles appears beneath this one (labelled 1 through however many RDI devices are configured) letting an individual device be silenced while RDI logging otherwise stays on.
- **Dimmer Control** — only shown when Enable QDC Lighting is on. Logs Zone 3+ preset changes (Zone 1–2 presets are logged under Outputs instead, since those are also reachable via CAI) and offset changes for every zone, including House/Stage.

- **NanoHost** — only shown when Enable NanoHost is on. Logs the Day Start/Day End/First Show End/Last Show Start triggers and the six top-level Status Flags.

Digital Projector Alternate Names

Four text fields, on the right-hand side of the Categories section — DP1 Power, DP1 Video, DP2 Power, DP2 Video — let you give each of the four Digital Projector flags (see Outputs Page) a different name specifically for Event Log messages. Leave any of them blank to keep the default label ("Dig Proj1 Power", etc.). These only change the text used when that flag's change is logged; they do not rename the buttons anywhere else in the plugin.

Recent Log Entries

A read-only history of the last 10 entries actually sent to the Core Manager Event Log, newest first, each with its own date/time stamp. This is a plugin-side convenience only — Core Manager's own Event Log remains the actual record — and it resets whenever the design (or the plugin script specifically) restarts.

Every entry sent, whether to Core Manager or this on-page history, is prefixed `eCNA-`*`n`*`: `, where *n* is the eCNA's Screen Id (see Screen Id) — useful for telling multiple eCNA units apart in a shared Core Manager Event Log. Before the Screen Id is first known (e.g. the very first connection attempt after a restart), *n* shows as `?`.

Control Pins

The following table lists the control pins available when this plugin is used in a Q-SYS schematic. All pins are optional; connect only those required by your design.

Pin Name	Value	String	Position	Direction
Setup				
IP Address		(text)		Input
TCP Port		(text)		Input
Connection Status	0 1 2 3 4 5	OK (Green) Compromised (Yellow) Fault (Red) Not Present (Grey) Missing (Orange) Initializing (Blue)	—	Output
Status LED	0–5	(same as Status)	—	Output
Connected	0 / 1	Off / On	0 / 1	Output
Reconnect	trigger	—	—	Input
Model		(text)		Output
Firmware Version		(text)		Output
Screen Id		(text — 0–63)		Output
Refresh Setup Data	trigger	—	—	Input
Program State				
Control State		IDL / RUN		Output
Stopped State		OK / STP / FLT / FIR		Output

Pin Name	Value	String	Position	Direction
Cue Number		(text)		Output
Program Number		(text)		Output
Segment Name		(text)		Output
Segment Index		(text)		Output
CAI CMD Access		OK / BLK		Output
Transport Commands				
Cmd Start	trigger	—	—	Input
Cmd Stop	trigger	—	—	Input
Cmd Cue	trigger	—	—	Input
Cmd Abort	trigger	—	—	Input
Cmd Fault	trigger	—	—	Input
Cmd Clear Fault	trigger	—	—	Input
Cmd Reset	trigger	—	—	Input
Cmd Cancel Alarm	trigger	—	—	Input
Set Program		(text / combo box, 1–9)		Input
Cmd Set Program	trigger	—	—	Input
Cmd Set And Run	trigger	—	—	Input
Macros (1 – Num Macros)				
Macros 1 … Macros N	trigger	—	—	Input
RDI (1 – Num RDI × 16)				
RDI 1 … RDI 80	trigger	—	—	Input
Outputs				
House Lights 1–5 (1=None 2=Down 3=Mid1 4=Mid2 5=Up)	0 / 1	Off / On	0 / 1	Input / Output
Stage Lights 1–3 (1=None 2=Down 3=Up)	0 / 1	Off / On	0 / 1	Input / Output
Curtain 1–3 (1=None 2=Close 3=Open)	0 / 1	Off / On	0 / 1	Input / Output
Masking 1–4 (1=None 2=Flat 3=Scope 4=Special)	0 / 1	Off / On	0 / 1	Input / Output
Lens 1–4 (1=None 2=Flat 3=Scope 4=Special)	0 / 1	Off / On	0 / 1	Input / Output
Sound 1–9 (1=None 2=Aux1 3=Aux2 4=Dig1 5=Dig2 6=Mono 7=NonSync 8=SR 9=SVA)	0 / 1	Off / On	0 / 1	Input / Output
Mute	0 / 1	Off / On	0 / 1	Input / Output
Dig1 Power	0 / 1	Off / On	0 / 1	Input / Output
Dig1 Video	0 / 1	Off / On	0 / 1	Input / Output
Dig2 Power	0 / 1	Off / On	0 / 1	Input / Output
Dig2 Video	0 / 1	Off / On	0 / 1	Input / Output
Aux Projector (Slide Projector)	0 / 1	Off / On	0 / 1	Input / Output
Aux Out 1–16	0 / 1	Off / On	0 / 1	Input / Output
Dig1 Power Name (Event Log alternate name, v1.4)		(text, blank = default)		Input

Pin Name	Value	String	Position	Direction
Dig1 Video Name (Event Log alternate name, v1.4)	(text, blank = default)			Input
Dig2 Power Name (Event Log alternate name, v1.4)	(text, blank = default)			Input
Dig2 Video Name (Event Log alternate name, v1.4)	(text, blank = default)			Input
Input & Output Flag Names				
In Names 1–16	(text — user-defined name for In1–In16)			Output
Out Names 1–16	(text — user-defined name for Out1–Out16)			Output
QDC-400 Dimmer Lighting (present only when Enable QDC Lighting is on)				
Zone Name 1 … Zone Name (QDC Zones)	(text — zone name, House/Stage/user-defined)			Output
Zone Preset 1 … Zone Preset (QDC Zones × 5) (5 per zone, in order Up/Down/Mid1/Mid2/None)	0 / 1	Off / On	0 / 1	Input / Output
Zone Offset 1 … Zone Offset (QDC Zones)	-100 – 100	—	—	Input / Output
Zone Offset Text 1 … Zone Offset Text (QDC Zones)	(text, editable, -100 to 100)			Input / Output
Zone Offset Inc 1 … Zone Offset Inc (QDC Zones)	0 / 1	Off / On	0 / 1	Input / Output
Zone Offset Dec 1 … Zone Offset Dec (QDC Zones)	0 / 1	Off / On	0 / 1	Input / Output
Channel Level 1 … Channel Level (QDC Channels)	0 – 100	—	—	Output
Channel Level Text 1 … Channel Level Text (QDC Channels)	(text, e.g. "75%")			Output
Channel Zone 1 … Channel Zone (QDC Channels)	(text — zone name this channel is assigned to)			Output
Channel Present 1 … Channel Present (QDC Channels)	0 / 1	Off / On	0 / 1	Output
QDC Board Count	(text — number of active QDC-400 boards found)			Output
nanoHost POS Management Data (present only when Enable NanoHost is on)				
NanoHost Ticket Sold	0 / 1	Off (Red) / On (Green)	0 / 1	Output
NanoHost In Show	0 / 1	Off (Red) / On (Green)	0 / 1	Output
NanoHost Pos Ok	0 / 1	Off (Red) / On (Green)	0 / 1	Output
NanoHost Pos Enabled	0 / 1	Off (Red) / On (Green)	0 / 1	Output
NanoHost Host Up	0 / 1	Off (Red) / On (Green)	0 / 1	Output
NanoHost Cancel (reverse logic)	0 / 1	Not Cancelled (Green) / Cancelled (Red)	0 / 1	Output
NanoHost Time Of Day	(text — live clock, generated locally)			Output
NanoHost Time Of Show (Feature Run Time)	(text, MM:SS)			Output
NanoHost Time Between Shows (Time Between Features)	(text, MM:SS)			Output
NanoHost Time Before Start (Time Before Feature Start)	(text, MM:SS, 0:00–255:59)			Output

Pin Name	Value	String	Position	Direction
NanoHost Time After Start (Time After Feature Start)	(text, MM:SS, 0:00–255:59)			Output
NanoHost Time Before End (Time Before Feature End)	(text, MM:SS, 0:00–255:59)			Output
NanoHost Time After End (Time After Feature End)	(text, MM:SS, 0:00–255:59)			Output
NanoHost Show RowNum 1–10	(text — row number, hidden with the rest of an empty row)			Output
NanoHost Show StartTime 1–10	(text — performance start date/time)			Output
NanoHost Show RunTime 1–10	(text — minutes, as reported)			Output
NanoHost Show PerfNum 1–10	(text — performance number)			Output
NanoHost Show Sold 1–10	0 / 1	Off (Red) / On (Green)	0 / 1	Output
NanoHost Show Cancel 1–10 (reverse logic)	0 / 1	Not Cancelled (Green) / Cancelled (Red)	0 / 1	Output
NanoHost Show Title 1–10	(text)			Output
NanoHost Show FS 1–10 (First Show Start)	0 / 1	Off / On	0 / 1	Output
NanoHost Show FE 1–10 (First Show End)	0 / 1	Off / On	0 / 1	Output
NanoHost Show LS 1–10 (Last Show Start)	0 / 1	Off / On	0 / 1	Output
NanoHost Show LE 1–10 (Last Show End)	0 / 1	Off / On	0 / 1	Output
NanoHost Show Start 1–10 (Cyan highlight, currently-active performance)	0 / 1	Off / On	0 / 1	Output
NanoHost Show End 1–10 (Yellow highlight, in progress past showtime)	0 / 1	Off / On	0 / 1	Output
NanoHost Day Start	0 / 1	Off (Grey) / On (Green)	0 / 1	Output
NanoHost Day Start Trigger	0 / 1	Off / On	0 / 1	Output
NanoHost Day Start Offset	-180 – 60	—	—	Input / Output
NanoHost Day Start Offset Text	(text, editable, -180 to 60)			Input / Output
NanoHost Day Start Offset Inc	0 / 1	Off / On	0 / 1	Input / Output
NanoHost Day Start Offset Dec	0 / 1	Off / On	0 / 1	Input / Output
NanoHost Day End	0 / 1	Off (Grey) / On (Green)	0 / 1	Output
NanoHost Day End Trigger	0 / 1	Off / On	0 / 1	Output
NanoHost Day End Offset	-60 – 240	—	—	Input / Output
NanoHost Day End Offset Text	(text, editable, -60 to 240)			Input / Output
NanoHost Day End Offset Inc	0 / 1	Off / On	0 / 1	Input / Output
NanoHost Day End Offset Dec	0 / 1	Off / On	0 / 1	Input / Output
NanoHost First Show End Trigger	0 / 1	Off / On	0 / 1	Output
NanoHost First Show End Offset	-60 – 60	—	—	Input / Output
NanoHost First Show End Offset Text	(text, editable, -60 to 60)			Input / Output
NanoHost First Show End Offset Inc	0 / 1	Off / On	0 / 1	Input / Output
NanoHost First Show End Offset Dec	0 / 1	Off / On	0 / 1	Input / Output

Pin Name	Value	String	Position	Direction
NanoHost Last Show Start	0 / 1	Off (Grey) / On (Green)	0 / 1	Output
NanoHost Last Show Start Trigger	0 / 1	Off / On	0 / 1	Output
NanoHost Last Show Start Offset	-90 – 60	—	—	Input / Output
NanoHost Last Show Start Offset Text	(text, editable, -90 to 60)			Input / Output
NanoHost Last Show Start Offset Inc	0 / 1	Off / On	0 / 1	Input / Output
NanoHost Last Show Start Offset Dec	0 / 1	Off / On	0 / 1	Input / Output
Event Logging (present only when Enable Event Log is on)				
Log Enable	0 / 1	Off / On	0 / 1	Input / Output
Log Enable Connection	0 / 1	Off / On	0 / 1	Input / Output
Log Enable Inputs	0 / 1	Off / On	0 / 1	Input / Output
Log Enable Outputs	0 / 1	Off / On	0 / 1	Input / Output
Log Enable Program	0 / 1	Off / On	0 / 1	Input / Output
Log Enable Macros	0 / 1	Off / On	0 / 1	Input / Output
Log Enable RDI	0 / 1	Off / On	0 / 1	Input / Output
Log Enable RDI Dev 1 … (Num RDI) (present only when Num RDI > 0)	0 / 1	Off / On	0 / 1	Input / Output
Log Enable Dimmer (present only when Enable QDC Lighting is on)	0 / 1	Off / On	0 / 1	Input / Output
Log Enable NanoHost (present only when Enable NanoHost is on)	0 / 1	Off / On	0 / 1	Input / Output
Log History 1–10	(text, read only — newest entry first, with date/time stamp)			Output

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